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An intelligent content provider based on students learning style to increase their engagement level and performance

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ABSTRACT

Learning trend has been shifted from a conventional way to a digital way in the form of E-learning, but it faces a high dropout ratio. Lack of engagement is one of the primary factors reported for this issue as the same type of course content is presented to learners despite their different background, knowledge and learning styles. Different researchers used adaptive learning techniques to increase students' performance and engagement, but it has not affected too much for reducing the dropout ratio. One of the major problems with these adaptive techniques is to focus on one thing. To target this issue we have proposed the framework that provides adaptive content to each student based on his learning dimensions and knowledge background as per Bloom's Taxonomy. The results of the experiment show a significant increase in students' engagement and performance.

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Adaptive content; E-learning; engagement; learning dimensions; Bloom's Taxonomy

1. Introduction

Approximately 13.5k courses have been offered in a massive open online course (MOOC) platforms that facilitate more than 110 million students (Shah, 2019). Despite this rapidly increasing growth in the number of online learners, there is a consistently high dropout rate of MOOC learners. Furthermore, the course completion rate is below 13% (De Freitas et al., 2015; Kim et al., 2018). The literature has identified three primary factors for the huge dropout rates: low engagement, non-interactive learning environment and difference in knowledge background (Kim et al., 2018; Kuzilek et al., 2015). Lack of student engagement is the significant factor that has large impact on MOOC. Student engagement means the degree of effort and interest that student shows in learning the content of a specific course (Lewis et al., 2011). It is the indicator of quality education and effective learning (Robinson & Hullinger, 2008). Researchers agree that student engagement is essential for success and in transforming higher educational institutions (Murray, 2018). Different research studies have employed collaborative learning, gamification and emotions in synchronous learning activities to engage the learners in eLearning environment (D'Errico et al., 2016; Hassan et al., 2019). Student engagement can be increased by delivering the content that is easily understandable and according to their cognitive learning style. Engagement can also be enhanced by designing questions in such a way that promotes critical thinking (Smith et al., 2007).

Evaluation of student thinking and knowledge is a great challenge, but there are existing theories that are used to analyze it. Bloom's Taxonomy is one of them to analyze the level of knowledge. There are six levels of Bloom's taxonomy: remembering, understanding, applying, analyzing, evaluating and creating. Frequent usage of bloom's taxonomy is a valuable tool to access student

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(Masapanta-Carrión & Velázquez-Iturbide, 2017). Assessment based on bloom's taxonomy provides significant results in the detection of student learning (Krouska et al., 2018).

In online courses the same types of course content and material are provided to students irrespective of their varying thinking level, educational background, performance and learning styles (El Guabassi et al., 2018; Karagiannis & Satratzemi, 2018). This is one of the major reasons of huge dropout; therefore, the research community has focused toward the adaptive learning.

Adaptive learning is a learning process where the content is presented based on the responses of an individual student. Adaptive learning plays a significant role for increasing student performance, efficiency and retention level (Kakish & Pollacia, 2018). The purpose of adaptive systems is to observe the student's progress in teaching process in such a way that student understands the course content easily (Chen et al., 2018). This seems to be a common challenge that if students solve easy problems, they might be less engaged and feel bored. Similarly, for difficult problems, their frustration level increases. Learning can be made adaptive by adapting gamification, feedback and learning styles (Dichev et al., 2014; Hassan et al., 2019; Karagiannis & Satratzemi, 2018)

The learning process is different for every learner as students learn according to their attitude and behavior (Akbulut & Cardak, 2012). They learn the material more quickly that suits their learning style (Brown, 2018). Adaptive learning positively affects students' motivation and performance (Karagiannis & Satratzemi, 2018). In traditional class rooms, it is very difficult for instructors to teach content according to the learner style and its performance due to limited time and setting of the class. However, the digital platforms have shifted learning trends from the conventional way to electronic versions that make the adaptive learning possible (Chang et al., 2013).

Learners in MOOCs are from all over the globe having different skills, subject knowledge, academic background and motivation. There is diversity in cognitive skills and knowledge, demands a different kind of learning for each student. However, the existing adaptive versions of E-learning platforms (Balasubramanian & Anouncia, 2018; Dichev et al., 2014; Kakish & Pollacia, 2018; Karagiannis & Satratzemi, 2018; Sweta & Lal, 2017) ignore this diversity and provide the same content to each learner based on their performance. Similar learning experience produces a feeling of boredom, high dropout rates, lack of interest and engagement (Balasubramanian & Anouncia, 2018; Onah et al., 2014). Research has been done to make the learning adaptive using learning style. These studies bring positive results on the remembering level and performance of students (Özyurt et al., 2014; Özyurt & Özyurt, 2015; Shaw, 2012), but do not improve motivation and engagement to such a level where course completion rate can be significantly improved. For significant outcomes, there is a need to design personalize adaptive content based on learning objective, learning style and performance. The innovation of this work is the Adaptive module that is based on knowledge Analyzer and Adaptive Content Module. Previously there is a very little work in which Adaptive module is designed in such a way to analyze knowledge and provide adaptive Content to individual students. We believe that a system with performance and adaptive content is required for better engagement and performance. The goal of this research is to build a framework which provides adaptive contents to increase student's performance and engagement level according to Bloom's Taxonomy.

More specifically, we have asked the following questions:

Does a system with adaptive content enhance student engagement?

Does student's performance increases as per blooms level by providing adaptive content?

Remaining paper is formulated in the following way. In Section II discusses existing studies relevant to adaptive learning. Section III lists down the methodology proposed for reducing the dropout and providing adaptive learning. Section IV contains the experiment section. Results and discussion is provided in section V. The paper concludes at section VI.

2. Literature review

2.1. Adaptive learning

Adaptive learning is a learning process in which learning content can be changed as per the need of an individual. The idea of adaptability in e-learning emerged when the researcher started working on personalised learning (Vélez et al., 2008). Theoretical approaches to make the learning adaptive are categorized into four areas: macro-adaptive; micro-adaptive approach; aptitude treatment instruction and constructive collaborative. The macro-adaptive approach emphasizes on learning objectives and aims to achieve learning outcomes by developing skills and knowledge (Mödritscher et al., 2004). Micro-adaptive approach analyzes interaction and ongoing learning tasks by giving instructions on learner's weak areas (Froschl, 2005). Aptitude treatment theory believes that every student needs instruction that matches with their personality (Lee & Park, 2008). Collaborative approach uses perception to focus on the performance of the learner. This approach gives independency to learner to improve their knowledge (De Corte, 2012). Adaptive research revolves on one of the above theoretical models while learning styles, gamification, performance and knowledge-based learning are the dimensions to make the learning adaptive.

2.2. Adaptive learning based on learning style

Learning style models used to classify students in a way how they process and perceive information. It defines how a student interacts in a learning environment. There are several models to classify the particular learning style of a student. These include Kolb's Learning Style Inventory (Kolb, 2007), Felder–Silverman Learning Style Model (Felder & Silverman, 1988) and Myers–Briggs Type Indicator (MBTI) (Myers et al., 1998). This section discusses the existing work that used one of the above models to identify the learning style. Karagiannis and Satratzemi (2018) proposed the mechanism for providing adaptive courses in Moodle by automatically detecting student's learning styles. They used a data-driven approach for identifying learning style with the help of index of learning style (ILS) questionnaire. The result indicates that students' performance and motivation increases by using adaptive mechanism. Othman and Amiruddin (2010) (Othman & Amiruddin, 2010) proposed a learning model VARK (Visual, Aural, Read/Write, Kinesthetic) that involves intelligence skills for information gathering and understanding. Balasubramanian and Anouncia (2018) (Balasubramanian & Anouncia, 2018) discussed a reinforcement model for adaptive learning environment based on the cognitive skills of the learners. These skills include memory, concentration, perception and logical thinking. The author first detects learning style based on cognitive skills, maps these skills into Bloom's Taxonomy with learning object and then creates a reinforcement model to keep track and provide feedback for competency level improvement. Their results showed 8% improvement in high level memory competency. Dorça et al. (2016) proposed rule-based expert system for the classification of learning objects using dynamic and automatically detecting learning style. Felder-Silverman model is used by them for detecting learning style (Dorça et al., 2016)

2.3. Adaptive learning based on gamification

Researchers also used gamification elements in eLearning to make the system adaptive and increase motivation of learners. Caro-Alvaro et al. (2017) suggested eLearning platform using gamification element points and rewards to engage and motivate learners using communication and queries (Caro-Alvaro et al., 2017). Kim & Lee proposed a dynamical model of game-based learning (Kim & Lee, 2015) that maximized education effectiveness; they used points, leaderboards, goals, badges, feedback and levels. Jagušt et al. (2018) presented the empirical study of three different types of gamified learning activities in lower primary mathematical class (Jagušt et al., 2018). The data logs were used to calculate performance level changes that were based on task completion rate. Higher performance level appeared in their study when gamified condition combined with

competition and adaptively with individual performance. An empirical study is presented by Kuo and Chuang (2016) for determining the motivation and engagement in online academic dissemination using gamification techniques (Kuo & Chuang, 2016). The result of their study reveals that their system motivates and retains users on the promotion of academic dissemination. Hassan et al. (2019) proposed a framework that provides adaptive gamification experience according to student learning (Hassan et al., 2019). Their proposed solution identifies learning style of students based on their interactions with the system and its reduced dropout ratio by 26%.

2.4. Adaptive based on content, feedback and performance

Learning objectives play a very vital role in effective educational process and desired learning outcome. These objectives are defined by the instructor that must be clear and meet the needs of the learner. The way the learning goals are defined is very important to help the students to understand what is supposed to be achieved. Revised Bloom's Taxonomy (RBT) is a framework for classification of learning objectives. The six levels of RBT are: remembering, understanding, applying, analyzing, evaluating and creating (Krathwohl, 2002). Researchers used this framework for course and assessment designing to make learning adaptive. Krouska et al. (2018) proposed computerized adaptive assessment based on revised Blooms' Taxonomy and their results provide a better detection in identifying learning gap (Krouska et al., 2018). To map the objectives and motivation in a course RBT is used by (Antony and joy) for descriptive analysis of students in their learning skill using data mining approach. Their methodology uses a classifier for classification of RBT level of question and achieves 98% accuracy in classifying RBT level of questions (Sami & Arumugam, 2020). (Zorluoğlu & Güven, 2020) analyzed that learning outcome and Exam questions are not properly as per revised Bloom's Taxonomy. 24% questions are those that do not belong to any learning outcomes. Their study suggests that teachers must know the learning level of the student. (Zorluoğlu & Güven, 2020).Chen et al. (2018) used the immediate corrective feedback for analyzing students' learning behavior in a blended learning environment (Chen et al., 2018). They identified the student performance patterns through checkable answers that showed the strong progress to those students who used feedback mechanism in their learning. Kakish and Pollacia (2018) describes adaptive learning and teaching techniques to increase students' retention level, scores and pass rate as well as the efficiency of faculty members (Kakish & Pollacia, 2018). Bright space D2L was also integrated with Learning Management System. By using the Simnet tool, student's progress was measured by the instructor. The analysis of their result showed 10 percent increase in student exam score and pass rate by using the adaptive learning techniques and decreasing the time spent during lecture preparation, grading and quizzes. Fuzzy logic approach based on Bloom's Taxonomy for the performance evaluation of e-learning is used by and their work shows that expert system based on Bloom's Taxonomy provides better analytical results to evaluate the student performance (Malatesh Akkur, 2019). El Guabassi et al. (2018) provides personalized content by considering learning style and context awareness. They tried to represent a link between learner profile and course material. Ahmed et al. (2018) suggested an adaptive model for open distance learning in which content is presented based on the performance of the student (Ahmed et al., 2018). Performance-based adaptive content increases satisfaction and decreases failure rate by 6.9%. Their research does not focus on student learning styles. Adaptive learning system-knowledge level (ALS-KL) based on the proficiency of a student is presented by Sfenrianto (Sfenrianto et al., 2018). Learner levels were improved significantly by 1.5 times in intermediate and 4 times in advanced level. Al-Nakhal and Abu-Naser (2017) build an adaptive intelligent tutoring system based on intelligent tutoring system builder (ITSB) by giving assistance and feedback to learner based on their behavior (Al-Nakhal & Abu-Naser, 2017). Their evaluation reveals the satisfaction result but is unable to engage students in such adaptive system.

In summary, although the available adaptive systems based on the performance, learning styles and knowledge level increase the satisfaction, performance and motivation of learners they are

Table 1. Comparison of studies employed in adaptive learning.

Research	Adaptive learning based on			Adaptive hybrid content based on Bloom's Taxonomy
	Student knowledge	Performance	Learning style	
Ahmed et al. (2018)	×	✓	×	×
El Guabassi et al. (2018)	×	✓	✓	×
Sfenrianto et al. (2018)	✓	✓	×	×
Karagiannis and Satratzemi (2018)	×	×	✓	×
Balasubramanian and Anouncia (2018)	✓	×	✓	×
Dorça et al. (2016)	×	×	✓	×
Our purposed system	✓	✓	✓	✓

unable to reduce the dropout rate. Students in the online learning platform have different background, knowledge, and learning styles. Due to this diversity every learner learns according to his capacity and skills. The existing studies lack in providing such type of personalized content to make the system adaptive that checks learner background, knowledge, and learning style and then present content based on his need and demand. Table 1 shows a comparison between the existing studies and our proposed work.

3. Proposed solution

In this paper, we have proposed a framework that not only helps instructors to create and develop adaptive content but also presents adaptive content to each student based on his learning dimensions, knowledge background and objectives. The overview of the proposed system is shown in Figure 1.

3.1. System functioning

We divide the functionality into two perspectives: student and teacher. With the help of learning type identifier, the learning style of each student is determined and assigned learning dimension. These learning styles are stored in learner model to construct learner's profile based on his identified learning style. Adaptive Module provides personalized content based on the prior knowledge. The adaptive module consists of two sub-modules: knowledge analyzer and content provider. The knowledge analyzer determines the knowledge level of learner using a performance-based assessment. The knowledge analyzer contains the question based on Bloom's cognitive domain level. These questions are retrieved from the assessment store where all the possible questions are stored and designed by the instructor. Adaptive Content module provides content to learner based on Bloom's Taxonomy level and his learning style using content-store that is designed by the instructor. The content-store contains content in three forms (text, audio and video). The contents provided by the proposed system are personalized according to the student's knowledge. The student is able to complete the course if he successfully progresses through the four levels (remembering, understanding, analyzing, and applying) of Bloom's Taxonomy. Hence the proposed system provides different learning experience to different students based on their learning style and learning objective.

3.2. System components

3.2.1. Learning type identifier

Learning type identifier identifies the learning style of each learner. Learning style detection can be achieved in two ways: collaborative approach and automatic approach. We are using collaborative

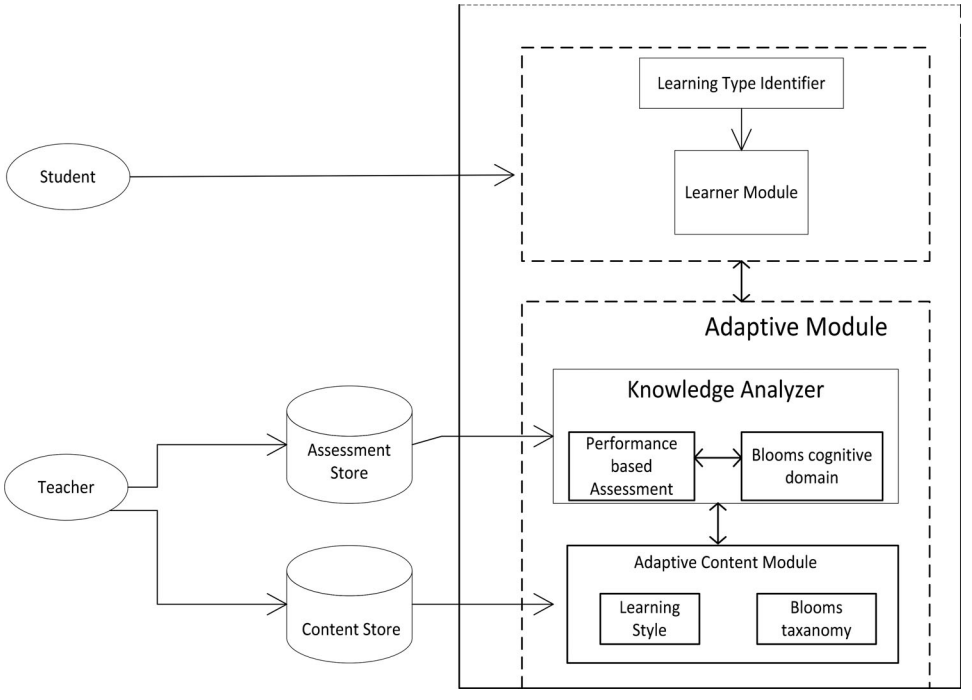


Figure 1. Architectural diagram of the proposed system.

approach which requires feedback from students regarding their learning preferences. There are different models of learning styles including VARK, Kolb, Felder-Silverman Learning Style Model and Myers-Briggs Type Indicator (MBTI) (Felder & Silverman, 1988; Kolb, 2007; Myers et al., 1998). Among these models, we are using Felder Silverman (ILS) as there is validity and reliability in a mature state and many researchers have used frequently in their research (Felder & Spurlin, 2005). Their model categorizes students into low, moderate and high preferences of learning style. Easy implementation, combination of different models and various preferences of a learner in single behavior are other supporting reasons for choosing it (Graf & Liu, 2010). Felder Silverman Learning Style (FSLS) categorizes the learner' style into four major dimensions and each dimension is further divided into two behaviors. These four dimensions and corresponding behavior are: Processing <Active, Reflective>, Perception <Sensing, Intuitive>, Organization <Sequential, Global>and Input<Visual, Verbal>. Table 2 represents the characteristics of FSLS dimensions.

3.2.2. Learner model

Learner model is a data store which stores student learning style. This module also logs the time spent by a student on each academic activity and his level of knowledge. We have used work of

Table 2. Learning behavior against each learning dimension.

Learning dimension	Learning behavior
Processing	Active: preferred learning by manipulation and work experience Reflective: preferred learning by thinking
Perception	Sensory: learning by concrete material and facts Intuitive: abstract material and conceptual view
Organization	Sequential: learning in linear steps Global: prefers a big picture and overall view
Input	Visual: learning from picture, diagrams Verbal: learning from words and display of text

Knutas et al. (2014) for content interaction to build the learner profile. Each time the student interacts with the system, a learning content is involved in the interaction. For example, if student reads, listens or watches a video, a learning object in the form of lecture having pdf, m4a or mp3 files is involved.

3.2.3. Adaptive module

Adaptive module has two submodules: knowledge analyzer and content provider. Both of the submodules are adaptive and finally provide adaptive content based on the learner knowledge, performance and learning style. The proposed adaptive module uses knowledge analyzer module to analyze the learner knowledge level and performance. Then, it provides adaptive contents based on Bloom's Taxonomy and learning style.

3.2.4. Knowledge analyzer

The standard evaluation of knowledge is the great challenge. However, the instructional design theories recognized Bloom's Taxonomy as a standard level of knowledge indicator (Koohang & Harman, 2007). Bloom's Taxonomy has been used by many researchers to represent the level of knowledge (Dorça et al., 2018; Krouska et al., 2018). This taxonomy was revised in 2001 and has six levels: remembering, understanding, applying, analyzing, evaluating and creating (Figure 2). We are using this taxonomy to analyze the knowledge at every level using a performance-based assessment module.

3.2.5. Performance-based assessment

This module is used to access the student performance against academic activities and provide adaptive content accordingly. This module uses an assessment store to determine the performance and knowledge of the student. Assessment store consists of questions that are defined by the instructor as per Bloom's Taxonomy. Learner's performance is measured up to the four levels of Bloom's Taxonomy (remembering, understanding, analyzing and applying) by this module as other two are high levels and are not included in this research work. Performance-based assessment store has high, average and low level of assessment that appears to student according to his performance. The diagram depicts the functionality of this module.



Figure 2. Bloom's taxonomy.

The assessment module evaluates the learner's remembering skills (first level of Bloom's Taxonomy). This judgment is based on performance in the Multiple Choice Questions (MCQs) quiz. These questions appear to student based on their learning style and performance. MCQs are also categorized as high, average and low level questions. At the start of the assessment, average level of question based on the learning type appeared to student. Next question either high level or low level appeared based on the result of the previous question. If the answer is true, then student has to answer high-level question or low level of question. The module also decides whether students have to move on the next level or they shall remain on the same level and need study more contents from the content store. Student moves to the next level if he correctly answers two questions. Similarly, the same approach is used to move at the next level. If the performance of learner is good, the number of questions appeared to him is fewer. On the other hand, if his performance is low, the number of questions is given to students is greater.

3.2.6. Adaptive content module

After analyzing knowledge and performance of learner, the module provides adaptive contents to the student based on their knowledge and learning style. The content-store that is defined by the instructor is used by this module to provide personalized content to learner. The content-store consists of contents to be provided for each learning style and knowledge level. Based on the assessment this module provides remembering, understanding, applying and analyzing level content and then moves to the assessment store. Every learner based on the learning style follows different paths to complete the course. Figure 3 shows the different paths that the students follow to complete the course.

4. Experiments

4.1. Experimental setup

The performance of the proposed system was evaluated with the help of two experiments. The participants of both experiments were the student of second-year undergraduate class in computer science. A total of 125 students were enrolled in the course. Data Structures were randomly distributed between two sections. Each section had both male and female students with the proportion of 47% and 53%, respectively having an average age of 21–26 years.

4.2. Experiment (I): without adaptive content

In experiment (I), sixty-five (65) students from section-A were enrolled in an E-learning course of Data Structures. For section A, we offered a system without adaptive content in which the content of the course is provided at start and after that, we assessed the student performance by using a quiz. In this system, there is no mechanism to analyze the knowledge of the learner. The system logged all the activities and time spent on solving the quiz by the student. The system provided the same content and quiz experience to all students irrespective of their performance and learning objective. The total marks with grades were displayed after the quiz completion.

4.3. Experiment (II): with adaptive content

In experiment (II), sixty (60) students from section (B) were enrolled in an online course of Data Structures. This course was offered with the proposed adaptive content framework which provides course content and quiz to students based on their performance and knowledge level. We use the data and logs of both sections to evaluate the system performance.

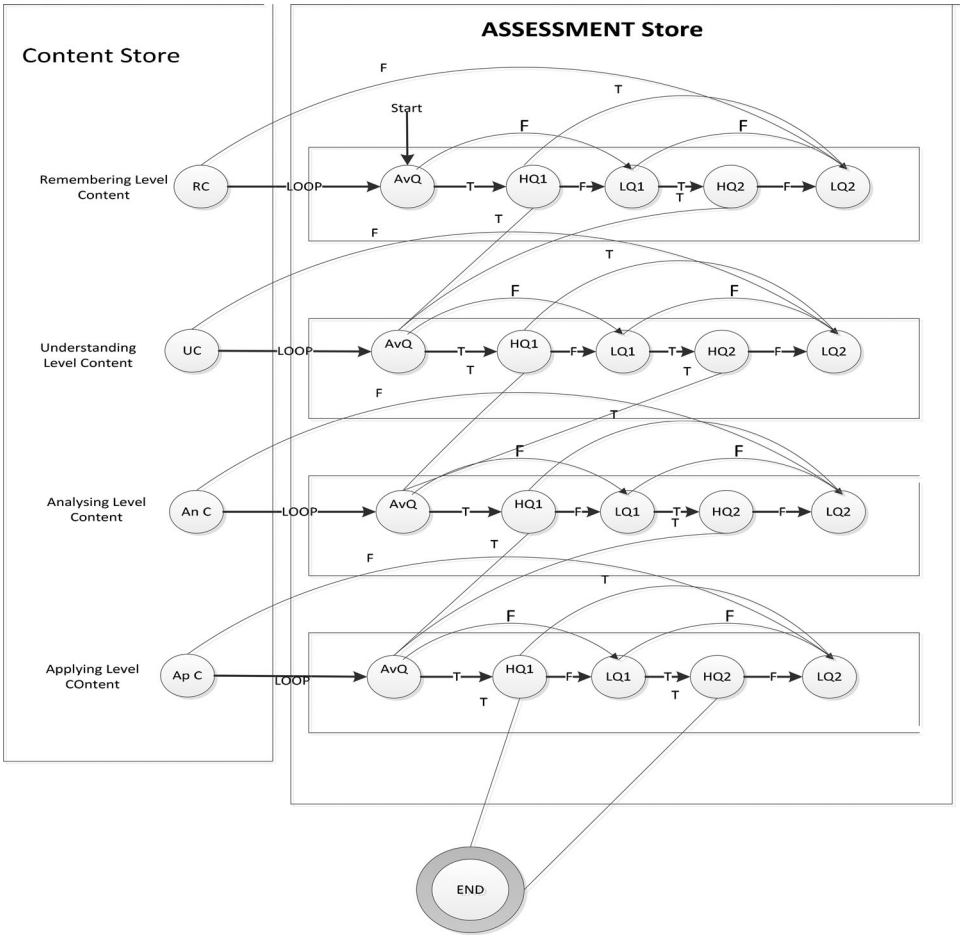


Figure 3. Sample assessment store.

5. Results

The system calculated the course completion rate when a window is displayed with completion status; otherwise, if student log outs, the system again starts where it left. The audio is marked completed when its play time is equal to the total audio time. However, for text contents, the system was relying on the student to click at the mark as a completed button. All the logs and time for all the interaction were saved to measure the engagement level. We used the effective engagement criteria of Kuh (Kuh, 2003) for measuring engagement level. Effective criteria of engagement relates to the time invested by a learner in their study. Results showed that students were more engaged and performance was increased in adaptive content as they directly reach the level at which they need improvement. Table 3 shows the Bloom's Taxonomy (remembering, understanding, analyzing and applying) level-wise content results. At remembering level, 12% students need to read the content as they are unable to solve remembering level assessments. The other 88% of student have a prior knowledge that's why they did not require remembering the level content. 33% of students require understanding level content and 67% have the prior knowledge. Similarly, 38% students face difficulty at analysis level; this is the level at which students face maximum level of difficulty. The ratio of students having prior knowledge at analysis level was 62%. At applying level 17% of students need to consult the required contents in order to clear the assessment. A total of 83% of students have successfully completed this level without requiring going to the

Table 3. Results based on Bloom's Taxonomy.

	Section B ($N = 58$) Level	Percentage for section B	Level-wise prior knowledge	Percentage of prior knowledge
Remembering	17	12%	41	88%
Understanding	29	33%	29	67%
Analyzing	36	38%	22	62%
Applying	12	17%	46	83%

content level stage. However, there is no segregation of level-wise content of section A that's why we keep the amount of time and the percentage of assessment questions for both sections to successfully compute the engagement level and performance.

Figure 4 shows the percentage of students who were asked to read the contents at different levels of Bloom's Taxonomy.

The maximum time spent by the students to understand the content in Table 4 was 20 h, whereas it was 11 h without adaptive content. Number of questions asked to each student is different in section A and section B, as represented in Figure 5. A total of 100% questions were asked in section (A) and they spend 5 h to complete the whole assessment, whereas 83% questions were asked in section B and their assessment completion time is 4.15 h.

6. Discussion

In this research, we have provided the adaptive content to each student for increasing engagement and performance. The proposed system automatically provides adaptive content to enrolled

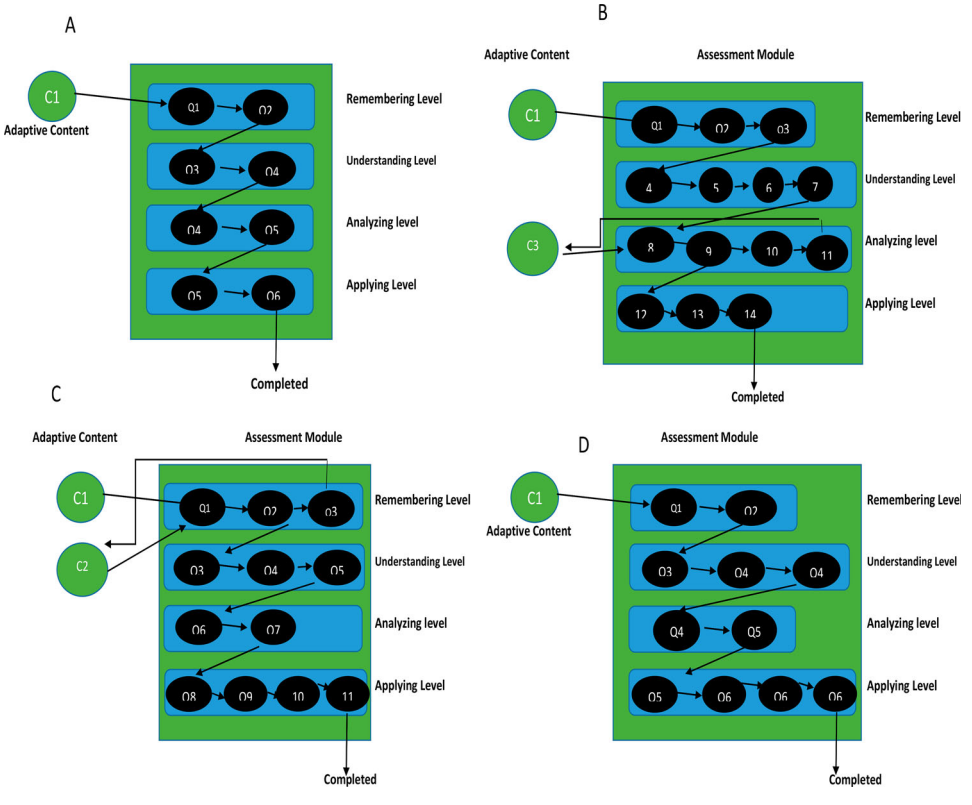


Figure 4. Path followed by different students.

Table 4. Comparisons of results from sections A and B.

Engagement	Without adaptive content	With adaptive content
Average hour spent per concept	11 h	20 h
Average hours spent on assessment	5 h	4.15 h
Percentage of questions appeared based on performance	100%	83%

students according to their performance, learning objectives and learning styles. Adaptive content enhances student's engagement and performance in e-learning. Our analysis from the result is that, at under-graduate level, most of the students have a prior knowledge of remembering and understanding level of content. They need Bloom's analysis level of content to be learned and assessed. if students get the contents that they alreayd knew, it shall be boring for most of the students and that may result in less engagement. The solution of our first research question addresses the issue by providing adaptive contents because, in this case, a student directly go to their required learning contents instead of wasting a time on the already knowing content. That is why a student spent more time and showed more engagement in learning a new concept using our adaptive system than to non-adaptive system. In our proposed system, a student can move to the next level of question only if he clears that level of assessment. If students are unable to clear the assessment, they move either the content or other assessment. So the performance criteria for our system are increased if student has fewer attempted assessments. It also address our second research question that the performance of students increased as they spent less time and attempted fewer number of assessments as compare to the non-adaptive system. The purpose of this research is to improve students' engagement and performance using adaptive content.

7. Conclusion and future work

To overcome the problem of dropout rates from online learning platforms, we have proposed a framework that provides personalized content to each student based on his performance and learning objectives which resulted in increased engagement and decreased dropout ratio. The results demonstrated that adaptive content, according to performance, learning dimensions and objectives, could significantly increase factors like performance and engagement in e-learning course. The research only uses the cognitive style (audio, text and video) of learning. Also, the system does not provide feedback or hint as used by Cohen (Chen et al., 2018) to think critically throughout the completion of the course. We work up to fourth levels of Bloom's Taxonomy. In future, the

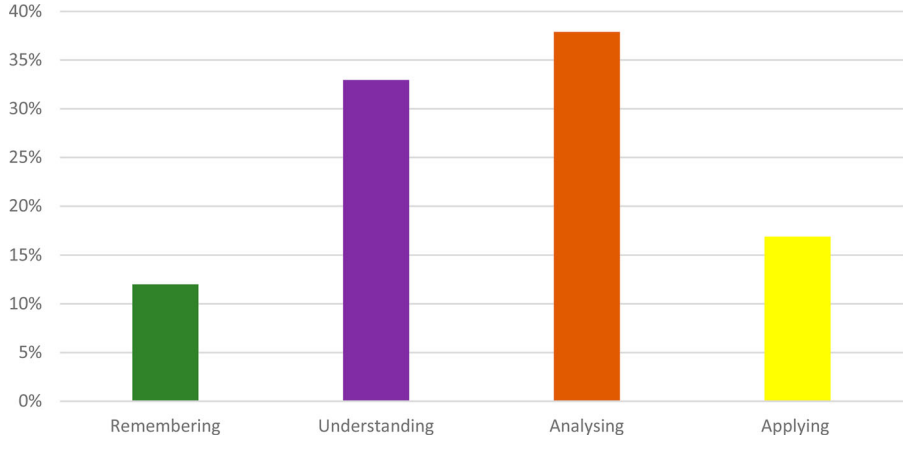


Figure 5. Level-wise results of Bloom's taxonomy.

system can be extended to provide feedback based on student performance and use interaction logs to identify the learner type of each student, so their performance should be improved.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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